OOP (C++): OO Modeling & Design

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Object Oriented Programming 2023/2024

1. Models
2. Modeling with UML
   - Class Diagram
3. (Some) Object Oriented Design Principles
   - The Single-Responsibility Principle (SRP)
   - The Open-Closed Principle (OCP)
   - The Liskov Substitution Principle
   - Dependency Inversion Principle
4. Conclusion

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A Kind of Definition

A model is an abstraction of the software system to be developed that makes the implementation (programming) of the system easier.

A model has three main characteristics (Herbert Stachowiak, 1973):

- *Mapping*: a model is always an image (mapping) of something.
- *Reduction*: a model does not capture all attributes of the original.
- *Pragmatism*: a model should be useful.

Quality of a Model

It is determined by the following characteristics (Bran Selic, 2003):

- Abstraction
- Understandability
- Accuracy
- Predictiveness
- Cost-effectiveness

UML

- To write a model, you need a modeling language.
- UML (Unified Modeling Language) is a language and modeling technique suitable for object-oriented programming UML is used to view, specify, build, and document object-oriented systems
- In this course we will use UML elements to explain the concepts and laws of OOP.
- There are free soft tools

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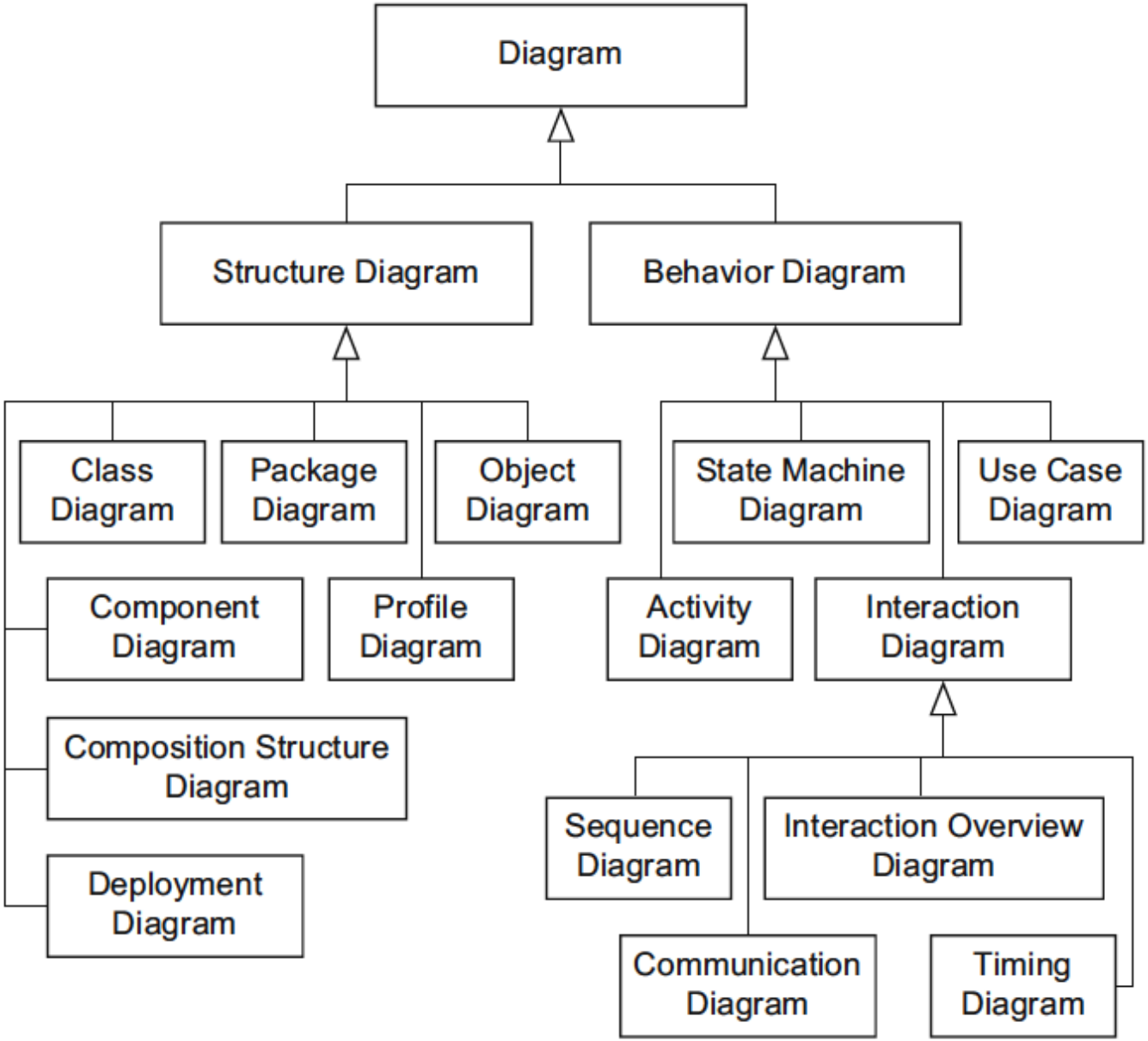
UML Diagrams



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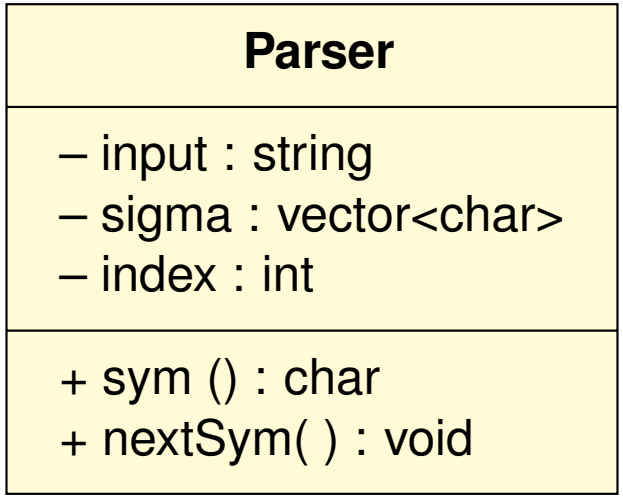
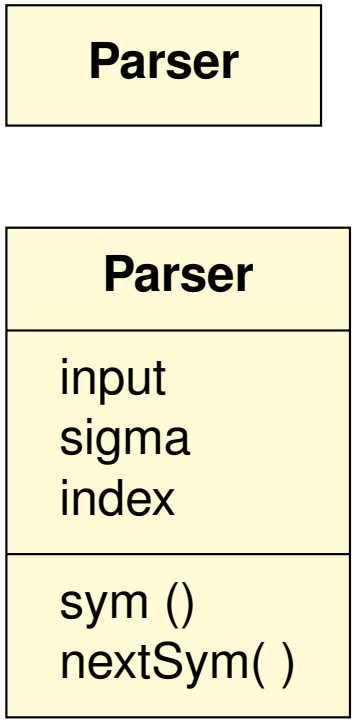
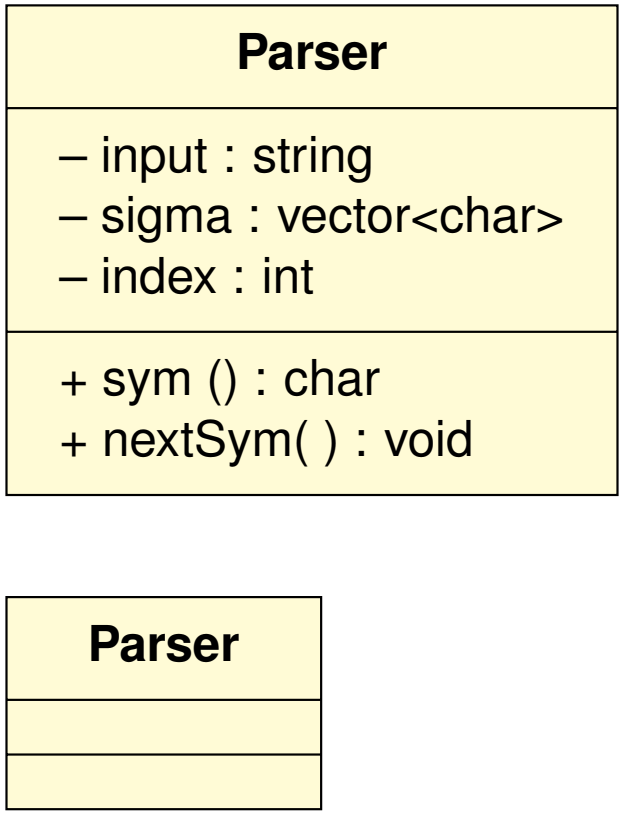
Specification of a Class in UML

```
class Parser {
 private:
  string input;
  vector<char> sigma;
  int index;
 public:
  char sym();
  void nextSym();
  ...
};
```



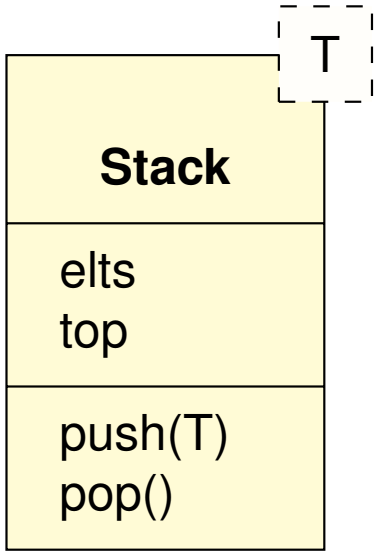
Notation


Parser
Parser
input
sigma
index
sym ()
nextSym( )


Parser
– input : string
– sigma : vector<char>
– index : int
+ sym () : char
+ nextSym( ) : void
Parser

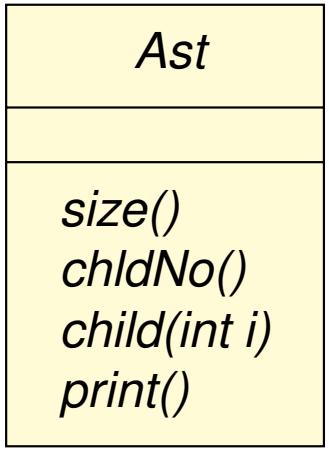
Parameterized Class

```
template<class T>
class Stack {
 private:
  T* elts;
  int top;
 public:
  void push(T);
  void pop();
  ...
};
```



Abstract Class

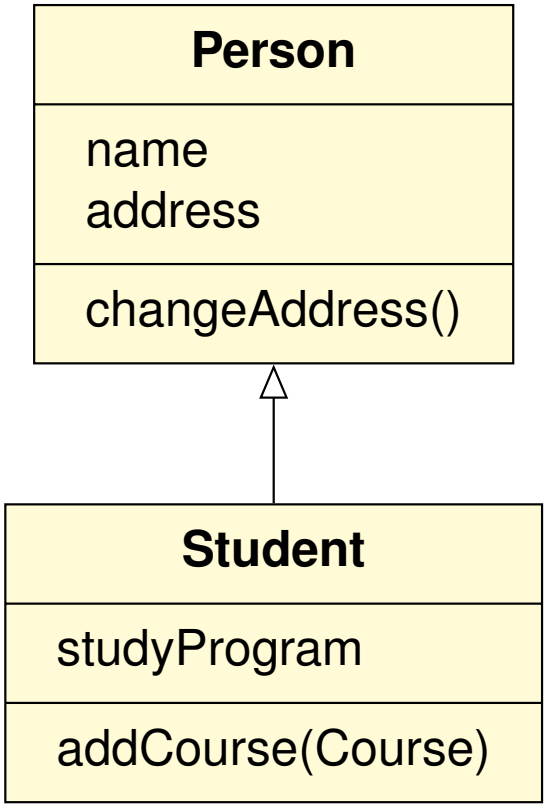
```
class Ast {
 public:
  virtual int size() = 0;
  virtual int chldNo() = 0;
  virtual Ast* child(int i) = 0;
  virtual void print() = 0;
};
```



Note the use of the italic font.

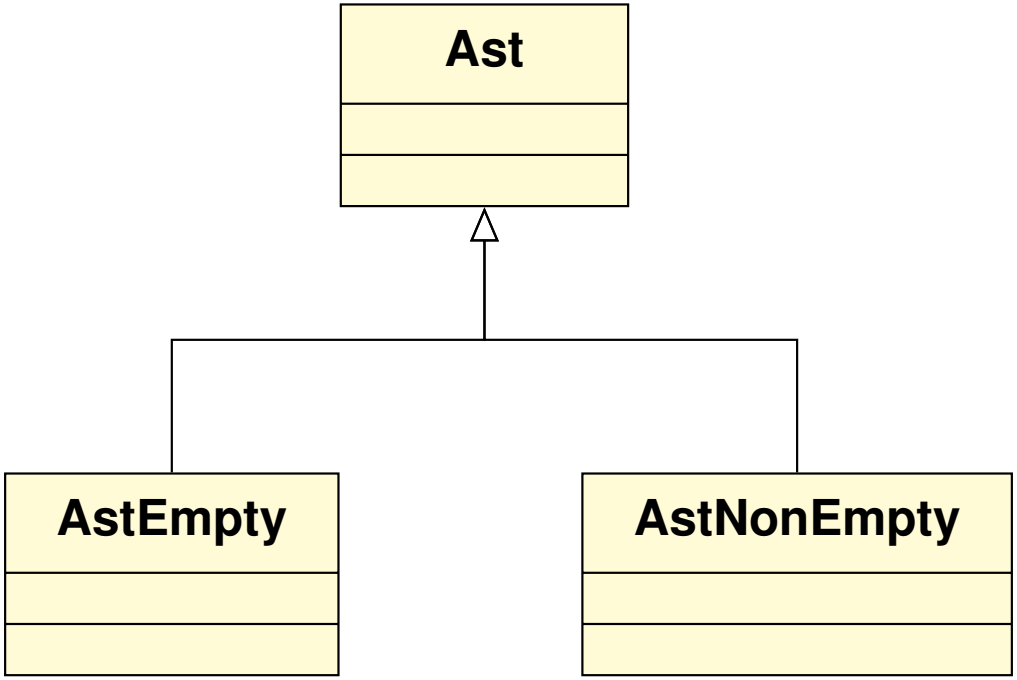
Generalization: Inheritance

```
class Person {
  string name;
  string address;
 public:
  void changeAddress();
};
class Student : public Person {
  list<Course> studyProgram;
 public:
  void addCourse(Course);
};
```



Generalization: Classification

```
class AstEmpty : public Ast {
  ...
};
class AstNonEmpty : public Ast {
  ...
};
```

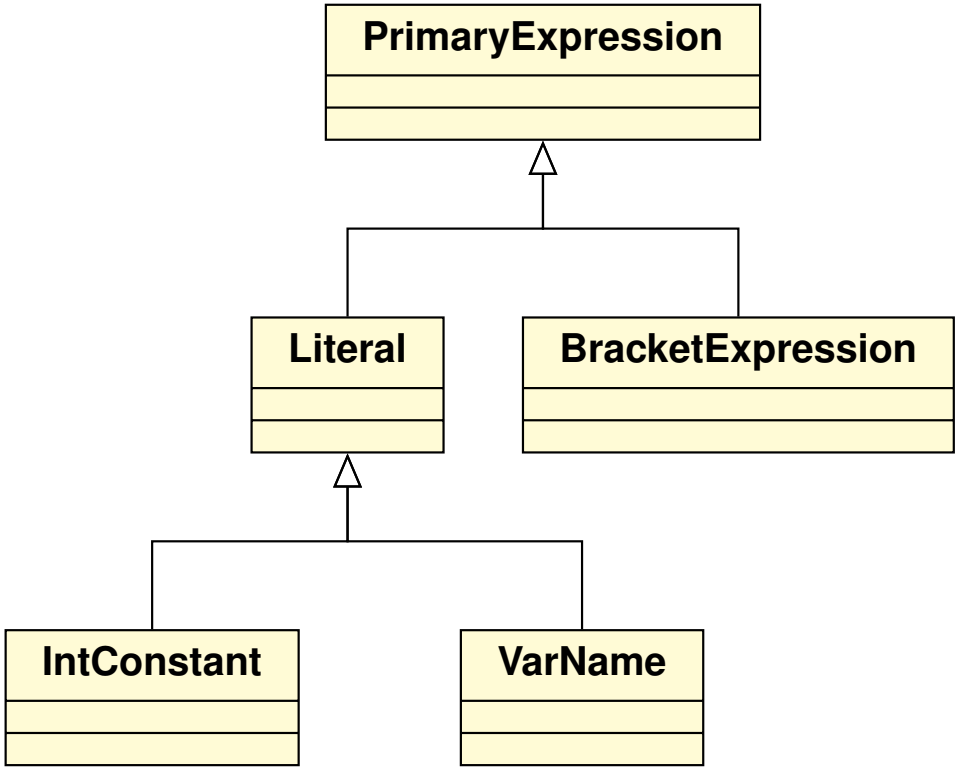


Classification: A Simple Case Study

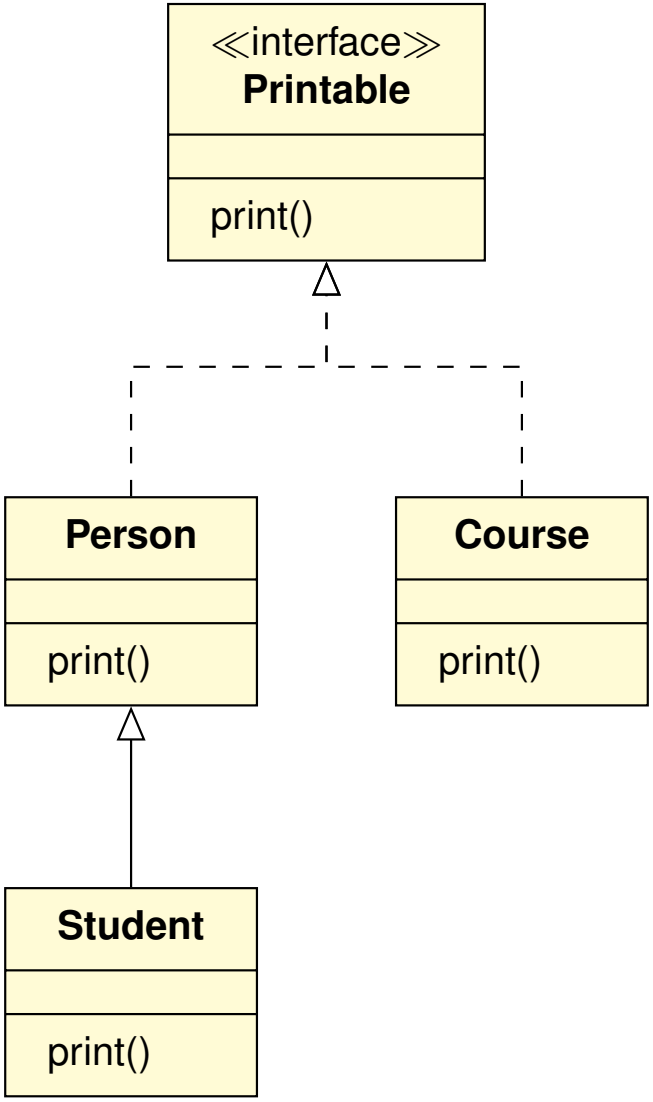
Frm C++ Grammar (simplified)

```
literal:
        int-constant
        var-name

primary-expression:
        literal
        ( expression )
```



Interface


≪interface≫
Printable
print()
Person
print()
Course
print()
Student
print()

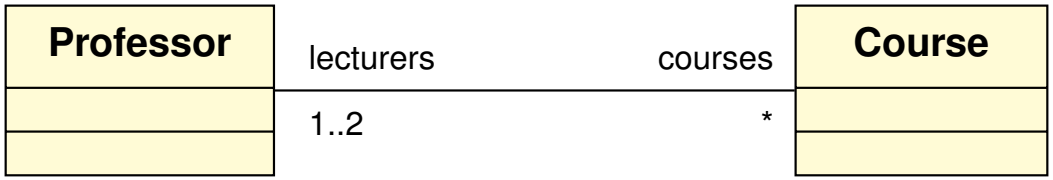
Interfaces in C++

Are described using abstract classes:

```
class Printable {
 public: virtual void print() = 0;
};

class Person : public Printable {
 public: virtual void print();
}
...
void Person::print() {
  ...
}
```

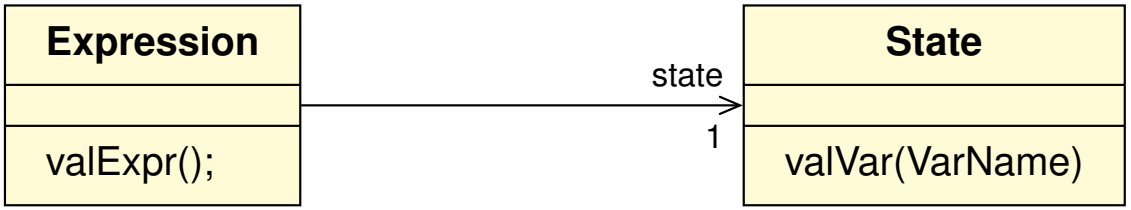
Bidirectional Binary Associations




```
class Professor {
 private: list<Course*> courses;
   ...
}
class Course {
  private: Profesor* lecturers[2];
   ..
}
```

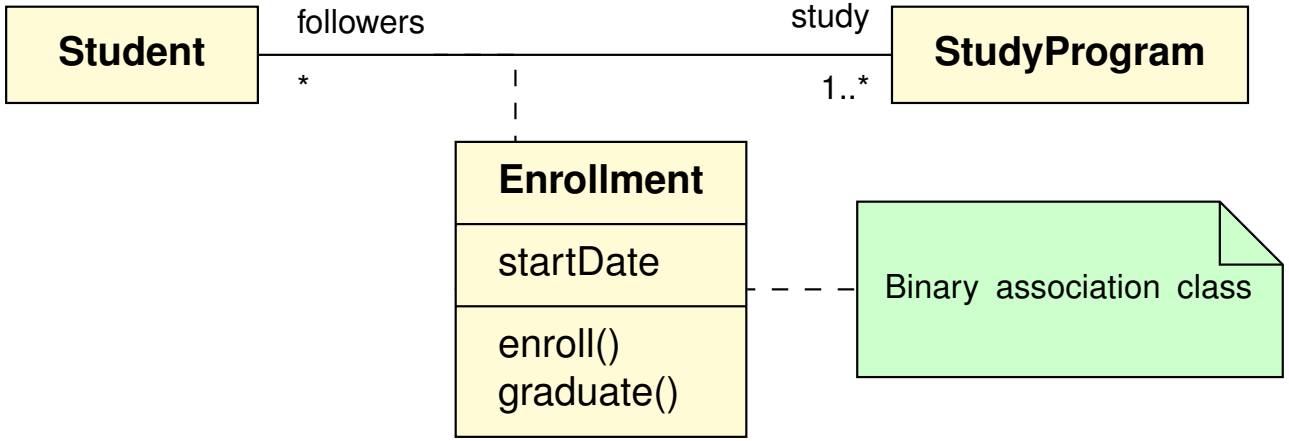
Unidirectional Binary Associations

– an expression is evaluated in a state (the expression knows the state it is evaluated)
– the state does not know for which expression it is used for evaluation



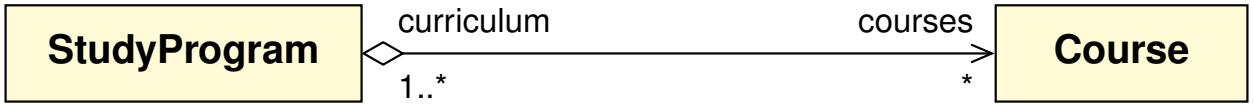

```
class Expression {
 private:  State* state;
 public: int valExpr(); // uses state->valVar()
   ...
}
class State {
  public : int valVar(VarName vn);
      // returns the value of the variable vn
   ...
}
```

Association Class




```
class Enrollment {
  // class invariant: study != empty
 private:
  list<Student*>  followers;
  list<StudyProgram*> study;
  Date startDate;
public:
  void enroll(Student, StudyProgram);
  void graduate(Student, StudyProgram);
  ...
};
```

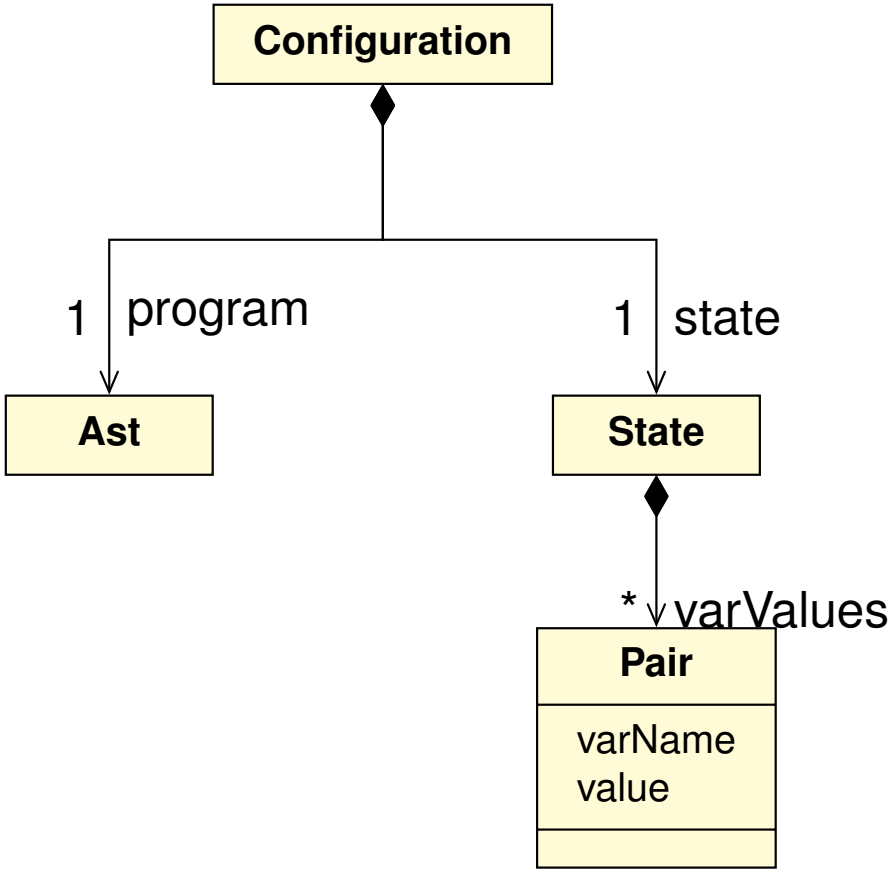
Shared Agregation




```
class StudyProgram {
 private: vector<Course*> courses;
 ...
};
class Course {
  // class invariant: includedIn != empty
 private: list<StudyProgram*> curriculum;
 ...
};
```

Strong Agregation (Composition)

configuration $= \langle$ program, state $\rangle$
state $=$ var1 $\mapsto$ *val*1, var2 $\mapsto$ *val*2, ...




```
class Configuration {
 private:
  Ast program;
  State state;
  ...
};
class State {
 private:
  map<VarName, Value> varValues;
 ...
};
```

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Definition

Single-Responsibility Principle

A class should have one and only one reason to change, meaning that a class should have only one job.

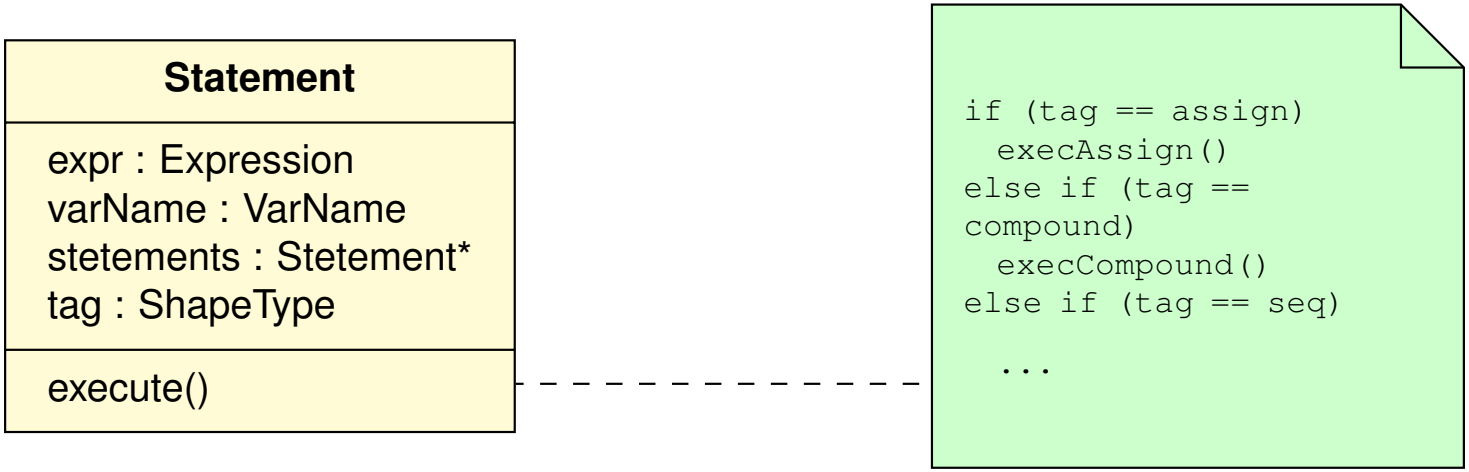
Robert C. Martin. Principles of Object Oriented Design.
https://drive.google.com/file/d/0ByOwmqah_nuGNHEtcU5OekdDMkk/view

~~SRP~~ 1/2

From C++ grammar (modified)

```
statement:
        var-name = expression ;  //assignment
        { statement }            // compound statement
        (statement)*             //sequential composition
        if ( expression ) statement   // if
        if ( expression ) statement else statement  //if
        ...
```

~~SRP~~ 2/2



Too many responsibilities.

SRP 1/2

From C++ grammar (simplified)

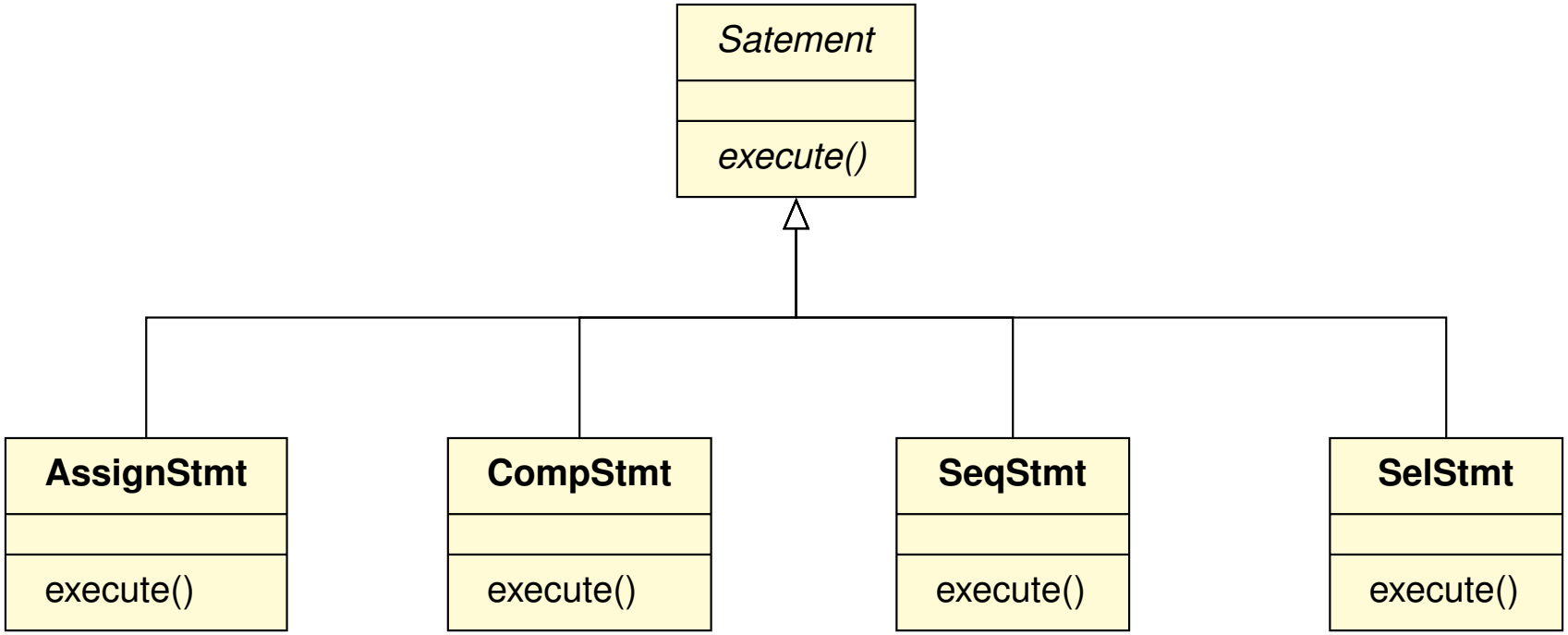
```
statement:
        assign-statement
        compound-statement
        statement-seq
        selection-statement

assign-statement:
        var-name = expression ;

compound-statement:
        { statement }
statement-seq:
        statement
        statement-seq statement

selection-statement:
        if ( condition ) statement
        if ( condition ) statement else statement
```

SRP 2/2


Satement
execute()
AssignStmt
execute()
CompStmt
execute()
SeqStmt
execute()
SelStmt
execute()

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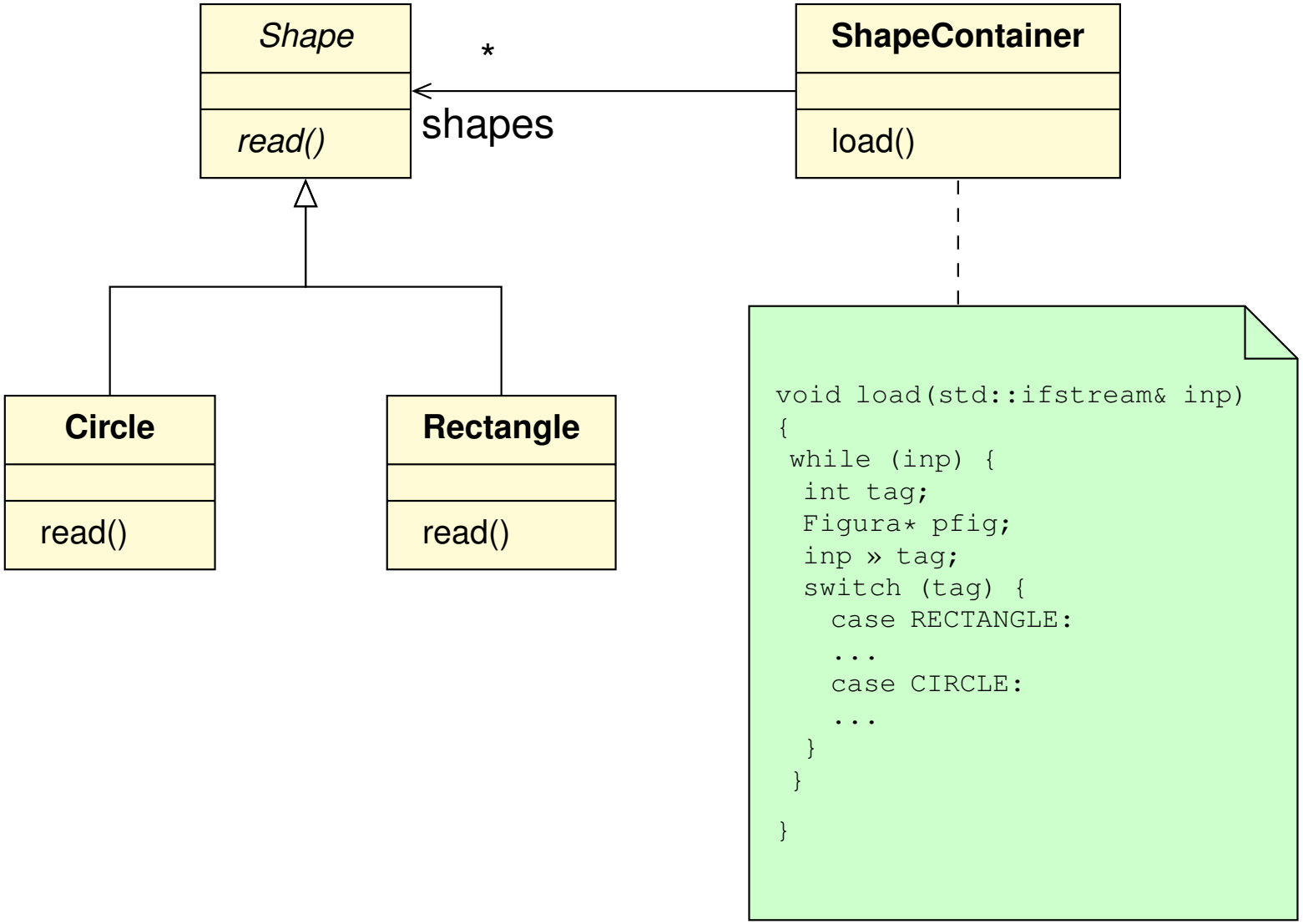
Definition

Open-Closed Principle

Software entities (classes, modules, functions, etc.) should be open for extension, but closed for modification.

Robert C. Martin. Principles of Object Oriented Design.
https://drive.google.com/file/d/0BwhCYaYDn8EgN2M5MTkwM2EtNWFkZC00ZTI3LWFjZTUtNTFhZGZiYmUzODc1/view

OCP



If we add more shapes, we have to modify `ShapeContainer::load()`.

OCP

The previous example can be fixed to respect OCP using Object Factory Pattern (next lecture).

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Definition

Liskov[1] Substitution Principle

"*Requirement that S is a subtype of T*:
Let $P(x)$ be a property provable about objects x of type T.
Then $P(y)$ should be true for objects y of type S, where S is a subtype of T."

In other words, S is a subtype of T if anywhere you can use a T, you could also use an S (a T can be substituted by an S).

NB Here the notion of subtype should be meant as behavioural subtype (we will explain later the difference).

Barbara Liskov, Jeannette M. Wing: A Behavioral Notion of Subtyping. ACM Trans. Program. Lang. Syst. 16(6): 1811-1841 (1994)

[1] Turing Award, 2008, for contributions to programming languages development, especially OO languages.

A Simple Example

```
class Base {
public:
    Base() = default;
    ~Base() = default;
    void doBaseAction() const { privateBaseAction(); }
private:
    void privateBaseAction() const { std::cout << "A base action\n"; }
};

class Derived : public Base {
public:
    Derived() = default;
    ~Derived() = default;
    void doDerivedAction() const { std::cout << "A Derived action.\n"; };
};

int main() {
    Base b;
    Derived d;
    d.doBaseAction(); // A base action
    return 0;
}
```

It seems that Derived is a (behavioral) subtype pf Base.

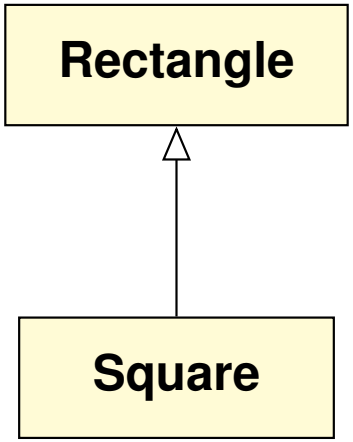
Another Simple Example

```
class Base {
public:
    Base() = default;
    ~Base() = default;
    virtual void doSomeAction() const { privateBaseAction(); }
private:
    void privateBaseAction() const { std::cout << "A base action\n"; }
};

class Derived : public Base {
public:
    Derived() = default;
    ~Derived() = default;
    void doSomeAction() const { std::cout << "A Derived but not Basic specific action.\n"; };
};

int main() {
    Base b;
    Derived d;
    d.doSomeAction(); // A Derived but not Basic specific action.
    return 0;
}
```

Now it seems that Derived is NOT a (behavioral) subtype pf Base! Overridden methods is an issue.

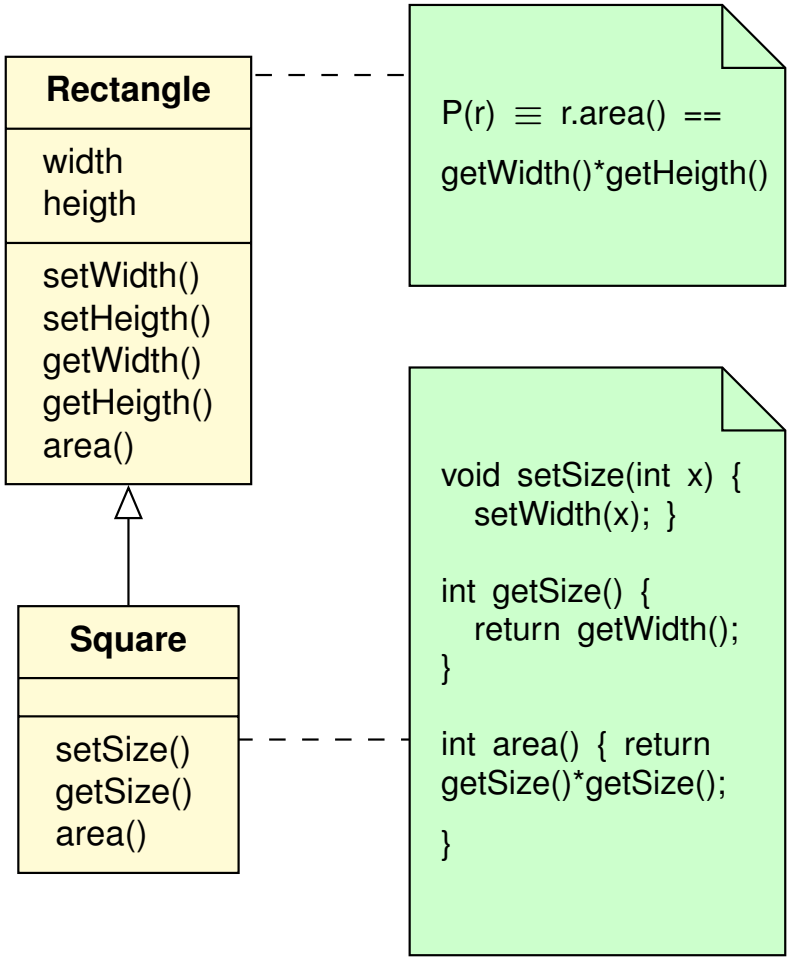
A Counter-Example 1/4

A rushed modeling of the relation "Any square is a rectangle" (this is an instance of the "is a" relation):



A Counter-Example 2/4

Let's refine it:



A Counter-Example 3/4

Consider the code:

```
Square s;
s.setWidth(5);        // inherited
s.setHeight(10);      // inherited
```

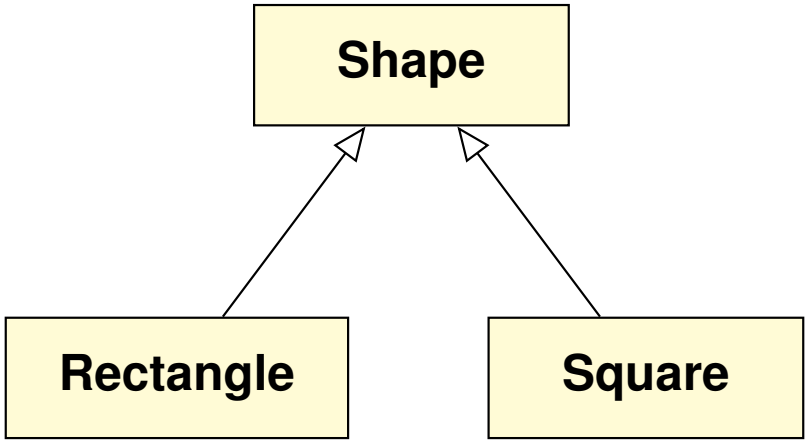
Assume that `s` plays the role of a `Rectangle`.
Does it satisfy the property P(`s`)? No.

Question

What is the right relationship between "subclass" and "(behavioural) subtype"?

A Counter-Example 4/4

A correct hierarchy:



Question

It seems that the subclass relation, given by

(child-class, partent-class)

is not the same with that of (behavioural) subtype.

So,
What is the right relationship between "subclass" and "(behavioural) subtype"?

Types, Types, Types, . . . 1/3

From (online) manuals[2] (partial):
Objects, references, functions including function template specializations, and expressions have a property called type, which both restricts the operations that are permitted for those entities and provides semantic meaning to the otherwise generic sequences of bits.

Type classification
The C++ type system consists of the following types:

- fundamental types
  - void
  - signed integer types int, long int, . . .
  - floating-point types float, double,. . .
  - . . .
- compound type
  - reference type
  - pointer type
  - array types
  - function types
  - class types
  - . . .

[2] https://en.cppreference.com/w/cpp/language/type

Types, Types, Types, . . . 2/3

So, a class is a type?

No, not completely true.

A class includes two orthogonal things:

- static/behavioral type
- implementation

Types, Types, Types, . . . 3/3

Static types (abstract view):

Declaration of X	Type associated to X	Remark
`int X;`	`int`	

`struct X {`
`  int a;`
`  double b;`
`}`
	`int` $\times$ `double`	product type

`int X(double a);`	`double` $\rightarrow$ `int`	function type

`class X {`
`  int size;`
`public:`
`  int getSize();`
`  void setSize(int a);`
`}`
	`void` $\rightarrow$ `X`
	`X` $\rightarrow$ `int`
	`X` $\times$ `int` $\rightarrow$ `X`
		constructor `X()`
		`getSize`
		`setSize`

For classes, only the types of the public members (interface) are considered.

Subtyping Problem 1/2

When a (static) type S is a subtype of a (static) type T (written $S <: T$)?

For instance, each `int` is a `long int`. Do we have

$$(\texttt{int}, _ + _ : \texttt{int} \times \texttt{int} \to \texttt{int})$$
$$<:$$
$$(\texttt{long int}, _ + _ : \texttt{long int} \times \texttt{long int} \to \texttt{long int})?$$

YES, if we consider $_ + _$ as partial functions (no over-/under-flow).

NO, if $_ + _$ as total functions (`MAX_INT` $+$ 2 on `int` is different from `MAX_INT` $+$ 2 on `long int`).

What happens if we model `int` and `long int` with classes?

Subtyping Problem 2/2

More general questions:

When $A \to B <: A' \to B'$?

If $typeOf(A)$ is the type of the class A, when

`class` B `:` `public class` A

implies

$$typeOf(B) <: typeOf(A)?$$

(i.e., when A and B respect LP?)

On Contravariance/Covariance/Invariance

Within a type system, the variance refers how the subtype relation is promoted to compound types.

- a rule is covariant if preserves the ordering of types $<:$
- a rule is contravariant if it reverses this ordering $<:$
- a rule is invariant if it is not neither covariant nor cotravariant (no subtyping relation can be derived)
- a rule is bivariant if it is both covariant and contravariant

E.g., the rule for function types is contravariant on arguments and covariant on result:

$$\frac{A' <: A \quad B <: B'}{(A \to B) <: (A' \to B')}$$

Contravariance/Covariance/Invariance in C++

C++ for overridden functions is invariant on parameters type and covariant on result type.[3]

From the manual:

The return type of an overriding function shall be either identical to the return type of the overridden function or covariant with the classes of the functions. If a function D::f overrides a function B::f, the return types of the functions are covariant if they satisfy the following criteria:

(8.1) — both are pointers to classes, both are lvalue references to classes, or both are rvalue references to classes

(8.2) — the class in the return type of B::f is the same class as the class in the return type of D::f, or is an unambiguous and accessible direct or indirect base class of the class in the return type of D::f

(8.3) — both pointers or references have the same cv-qualification and the class type in the return type of D::f has the same cv-qualification as or less cv-qualification than the class type in the return type of B::f.

[3] See also Course 5.

C++: Return Covariant Example 1

```
class Base
{
public:
    Base() = default;
    virtual Base* clone() const { return new Base; }
    virtual ~Base(){ std::cout << "~Base()\n"; }
    virtual void introduceYourself() const {std::cout << "I am a Base.";}
};

class Derived : public Base
{
public:
    Derived () = default;
    Derived* clone() const override { return new Derived; }
    virtual void introduceYourself() const override {std::cout << "I am a Derived.\n";}
    ~Derived() override { std::cout << "~Derived()\n"; }
};

int main() {
    Derived d;
    Base* pb = d.clone();
    pb->introduceYourself(); // I am a Derived.
    delete pb;
    return 0;
}
```

The method `clone` is covariant on result type.

C++: Return Covariant Example 2[4]

```
class Animal {
public:
    virtual void makeNoise() = 0;
    virtual ~Animal() =  default;
};

class Dog : public Animal {
public:
    void makeNoise() override { std::cout << "ham!\n"; }
};

class AnimalBreeder {
public:
    virtual Animal* produce() = 0;
};

class DogBreeder : public AnimalBreeder {
public:
     Dog* produce() override { return new Dog(); }
};

int main(){
    DogBreeder db;
    Dog* dp = db.produce();
    dp->makeNoise();  // ham
    delete dp;
}
```

The method `produce` is covariant on result type.

[4] The example is inspired from here: https://quuxplusone.github.io/blog/2019/01/20/covariance-and-contravariance/

C++: Argument Contravariant (Counter)Example

```
class Animal {
public:
    virtual void makeNoise() = 0;
    virtual ~Animal() =  default;
};

class Dog : public Animal {
public:
    void makeNoise() override { std::cout << "ham!\n"; }
};

class DogDoctor  {
public:
    virtual void treat(Dog *dog) { std::cout << "Dog treated!\n"; }
};

class AnimalDoctor : public DogDoctor {
public:
    void treat(Animal * animal) override { std::cout << "Animal treated!\n"; }  // error
};
```

Remember that C++ is invariant on arguments:

"If a virtual member function vf is declared in a class Base and in a class Derived, derived directly or indirectly from Base, a member function vf with the same name, parameter-type-list (9.3.3.5), cv-qualification, and ref-qualifier (or absence of same) as Base::vf is declared, then Derived::vf overrides Base::vf."

Behavioral Type of a Class 1/3

Consider the following simple example:

```
class Square {
  int size;
 public:
  Square(int s = 0);
  int getSize();
  int area() noexcept (false);
  void halve();
}
```

Behavioral Type of a Class 2/3

Includes also the behavioral interface of the objects and consists of:

- a description of the value space
  - class invariants
    Example: `getSize()` ≥ 0
  - for each constructor *c*
    - its signature
      Example: `Square` : `int` $\rightarrow$ `Square`
    - precondition
      Example: `s` $\geq$ `0`
    - postcondition
      Example: `getSize()` ≥ 0
    - the exceptions status

Behavioral Type of a Class 3/3

- for each public method *m*:
  - its signature
    Example: `area` : `Square` $\rightarrow$ `int`
    `halve` : `Square` $\rightarrow$ `Square`
  - precondition
    Example: `even`(`getSize`()) (for `halve`)
  - postcondition
    Example: `area`() == `0.25` * `old`(`area`()) (for `halve`)
  - the exceptions status
- for each public attribute *a*:
  - its type

A Possible Specification for the Behavioral Type

```
type Square {
   invariant getSize() >= 0
   Square(int s = 0)
     requires s >= 0
     ensures getSize() >= 0
     noexcept(true);
   int getSize()
     requires getSize() >= 0
     ensures getSize() >= 0
     noexcept(true);
  int area()
     requires true
     ensures result == getSize()*getSize()
     throw(overflow_error)
  void halve();
     requires even(getSize())
     ensures area() == 0.25*old(area())
     noexcept(true);
}
```

The Substitution Principle Explained using Behavioral Types 1/2

A child class (subclass) B is a behavioral subtype of the parent class (superclass) A, $behTypeOf(B) <: behTypeOf(A)$, if:

- the invariants of the superclass are preserved by the subclass (B values are A values): the conjunction of the B invariants implies conjunction of the A invariants, i.e.,

$$\bigwedge_{\phi \in Inv(B)} \phi \implies \bigwedge_{\phi \in Inv(A)} \phi$$

- for each overridden method m
  - $B::m()$ and $A::m()$ have the same number of arguments
  - contravariance of arguments: if the i-th argument of $B::m()$ is β_i and the i-th argument of $A::m()$ is α_i, then $\alpha_i <: \beta_i$
  - covariance of result: if the result type of $A::m()$ is α and the result type of $B::m()$ is β, then $\beta <: \alpha$

The Substitution Principle Explained using Behavioral Types 2/2

- exception rule: the exceptions status of $B::m()$ is the same with the exceptions status of $A::m()$
- semantics of the (overridden) methods is preserved:
  - the precondition of $A::m()$ implies the precondition of $B::m()$
  - the postcondition of $B::m()$ implies the postcondition of $A::m()$

N.B. This is a simplified version of the original definition.
Exercise. Show that $T <: T$ ($<:$ is reflexive).

The Substitution Principle Explained using "Design by Contract Methodology"

Design by Contract Methodology (Bertrand Meyer, 1986[5]):
"The interface specifications for software components should be defined in a formal, precise, and verifiable way."

Constraints on how contracts can interact with inheritance:

- For object states:

  Class invariants must be preserved in the subtype.
- For overridden methods:

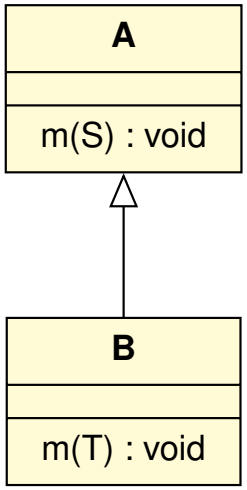
  Preconditions cannot be strengthened in the subtype.

  Postconditions cannot be weakened in the subtype.

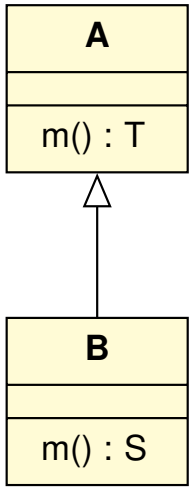
[5] Meyer, Bertrand: Design by Contract, Technical Report TR-EI-12/CO, Interactive Software Engineering Inc., 1986

Contravariance/Covariance Rule with Diagrams

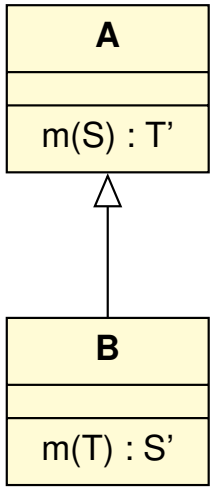
S <: T
S' <: T'



Contravariance of arguments

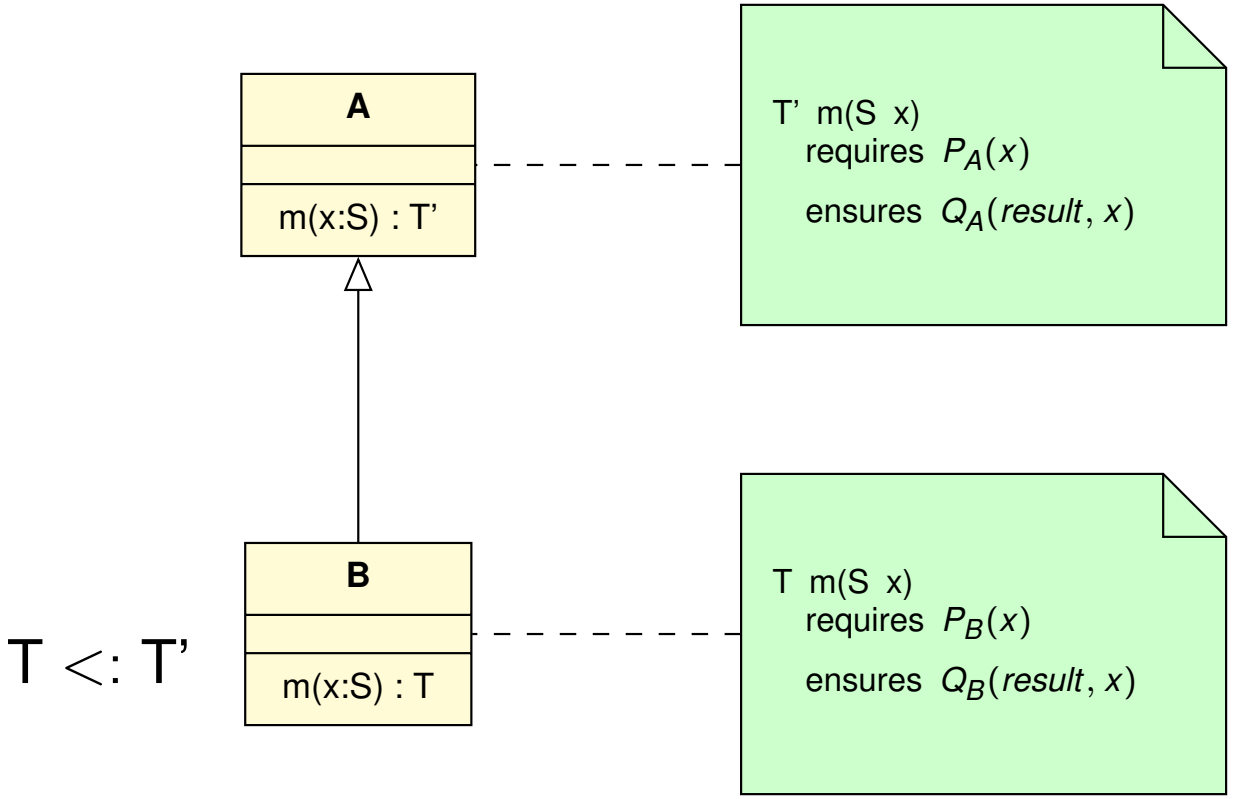


Covariance of result



Contravariance of arguments and Covariance of result

Semantics Rule with Diagrams



$$P_A \implies P_B$$
$$Q_B \implies Q_A$$

Contra-Example Revisited

The example "any square is a rectangle" violates several rules:

- the nethod `Square::setHeigth()` does not preserve the `Square` invariant `getWidth()` = `getHeigth()`
- the postcondition of `Square::area()` does not imply the postcondition of `Rectangle::area()`

Plan

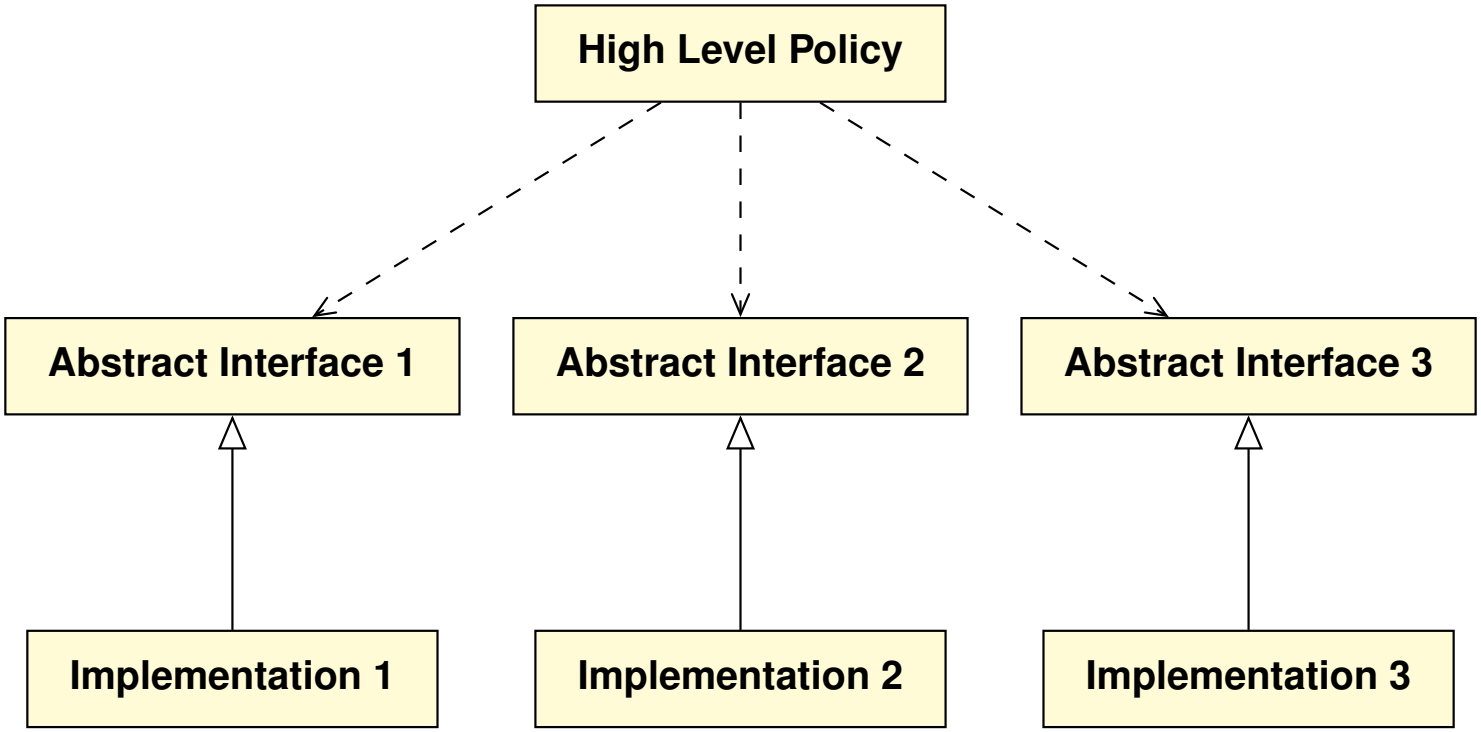
1. Models
2. Modeling with UML
   - Class Diagram
3. (Some) Object Oriented Design Principles
   - The Single-Responsibility Principle (SRP)
   - The Open-Closed Principle (OCP)
   - The Liskov Substitution Principle
   - Dependency Inversion Principle
4. Conclusion

Definition

- High-level modules should not depend on low-level modules. Both should depend on abstractions.
- Abstractions should not depend on details. Details should depend on abstractions.

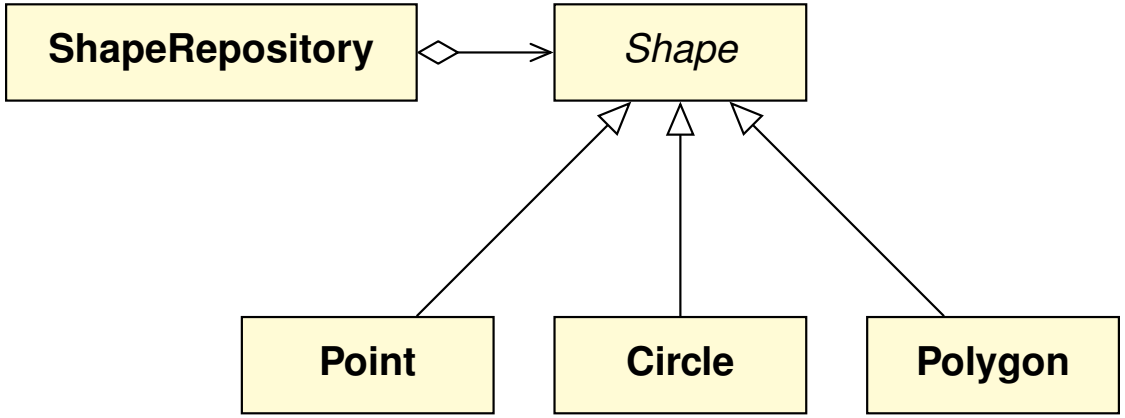
Robert C. Martin. Principles of Object Oriented Design.
https://fi.ort.edu.uy/innovaportal/file/2032/1/design_principles.pdf

Abstract View

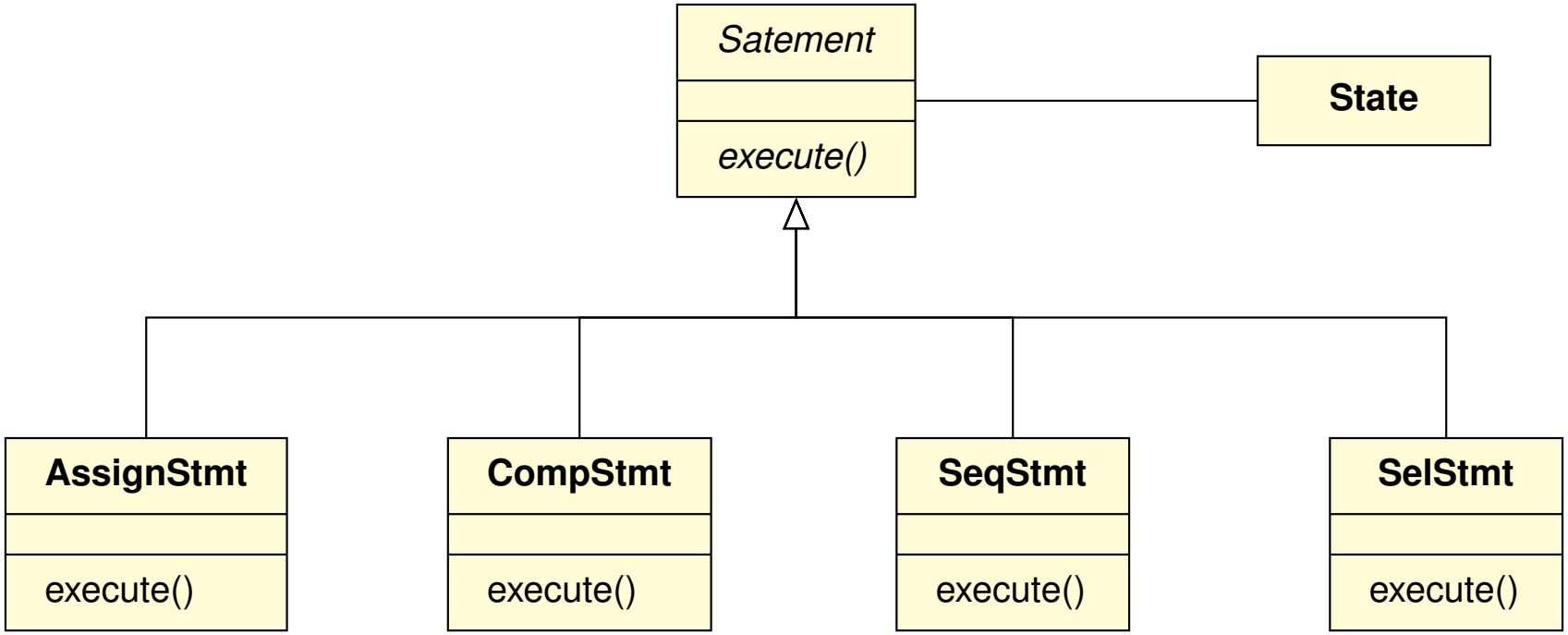


Note the notational difference between the dependence and inheritance relations. Inheritance also implies dependency.

Example



Example


Satement
execute()
State
AssignStmt
execute()
CompStmt
execute()
SeqStmt
execute()
SelStmt
execute()

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Conclusion

- Models first.
- Apply the right OO Design Principle wherever such a principle is applicable
- Revise your applications from previous laboratories and design their models and identify the places where principles are applicable.